Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

syome@uwyo.edu Laramie, WY, USA seungminyome

Education

University of Wyoming, Energy & Petroleum Engineering — Laramie, WY, USA, 2025 – present

- Advisor: Dr. Soheil Saraji
- Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam
- Coursework: Rock Mechanics, Enhanced Oil Recovery, Graduate English Writing, Geostatistics, Python for Engineers

Pusan National University, Naval Architecture & Ocean Engineering — South Korea, 2018 – 2024

- Advisor: Dr. Inwon Lee
- Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD

Research Experience

University of Wyoming, Graduate Research Assistant — Laramie, WY, USA, 2025 – present

- Developing a coupled solid–fluid simulation framework in solids4Foam (OpenFOAM-based) to model proppant grain behavior under compressive loading between parallel rock fracture walls
- Implementing J2 perfect-plasticity rock constitutive models and penalty-based contact algorithms to capture proppant embedment and fracture wall deformation
- Constructing structured O-grid meshes for proppant–rock contact problems; validated stress distributions against Hertz contact theory
- Performing laboratory training on triaxial cell operation, confining/pore-pressure control, and rock compressibility (bulk modulus) testing using DCI equipment

KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time) — South Korea, 2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Investigated gravity–capillary wave dynamics on elastic silicone films subjected to continuous air-blow loading above water
- Derived dispersion relations for gas–solid–liquid interface interactions under nonlinear elastic coupling
- Developed 2D ventilated supercavitation models and introduced wake-theory-based modifications, significantly improving agreement between theoretical predictions and experimental data for cavitator-driven cavity asymmetry
- Conducted vortex–wave interaction CFD simulations near free surfaces using ANSYS Fluent; analyzed free-surface-induced von Kármán vortex asymmetry mechanisms

Pusan National University — Department of Naval Architecture & Ocean Engineering, Undergraduate Research Assistant — South Korea, 2022 – 2024

- Performed RANS CFD validation of KCS hull resistance (Gothenburg 2010 Workshop benchmark) using STAR-CCM+; 1.16% error in total resistance coefficient CT; contributions: structured mesh optimization and post-processing
- Performed particle image velocimetry (PIV), hull resistance testing, and propulsion system design per ITTC 1957/1978 standards
- Developed ANN-based defect detection models using vibration and acoustic sensor data for ship structural health monitoring (MATLAB & Python)

Industry Experience

Samsung Heavy Industries Laboratory, Mechanical Engineer Intern — South Korea, 2024 – 2024

- Trained and evaluated artificial neural network architectures for Green Nuri vessel defect classification using MATLAB and Python

Pusan National University, Capstone Design Project — South Korea, 2023 – 2023

- Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria
- Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability

Publications

Computational analysis of proppant embedment and fracture conductivity in deformable rock using solids4Foam — in preparation
Seungmin Yome, Soheil Saraji
(Computers and Geotechnics)

Numerical Recovery of Hydroelastic Wave Dispersion in Coupled Elastic Film–Liquid Systems Using a Finite-Volume Fluid–Structure Interaction Solver — in preparation
Seungmin Yome
(Journal of Fluids and Structures)

Grants & Fellowships

Graduate Research Assistant (Full-funded Graduate Research Assistantship) — 2025–Present

- University of Wyoming

Research Scholarship — Jan 2024

- Korea Advanced Institute of Science and Technology

Undergraduate Research Student Grants — 2022–2024

- Pusan National University

Academic Excellence Scholarship — 2022–2024

- Pusan National University

Foreign Language Special Lecture Consignment Program Scholarship — Jan 2021

- Pusan National University

Creative Human Resources Development Project Group Scholarship — Jan 2018

- Pusan National University

Selected Activities & Awards

Invited to the Samsung Heavy Industries Wind Propulsion Ship Forum with travel award — July 2024

- Samsung Heavy Industries

Participating in the program ‘Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship’ — Jan. 2024

- Korean Society of Shipbuilding and Marine Engineering / Coast Guard

Invited to the Hyundai Heavy Industries Eco-Friendly Smart Ship Forum with travel award — Apr. 2023

- Hyundai Heavy Industries

TPC (Test Preparation Course) Certification — English Consignment Program July 2021

- Pusan National University

Teaching & Tutoring

Private Tutoring (Educational Volunteering), Mathematics Instructor South Korea

- Taught high school mathematics to undergraduate-level students as part of educational volunteering activities over two years 2022 – 2024

Private Tutoring, Physics & Chemistry Instructor — College Entrance Preparation South Korea

- Taught physics and chemistry to 10 high school seniors (Grade 12) preparing for the Korean college entrance exam (CSAT) over a 6-month intensive period 2023 – 2024
- 9 out of 10 students (90%+) gained admission to top-tier Korean universities

Skills

Simulation & CAE

Programming

Design & Engineering Tools

Laboratory

Languages

Korean
Native

English
Fluent